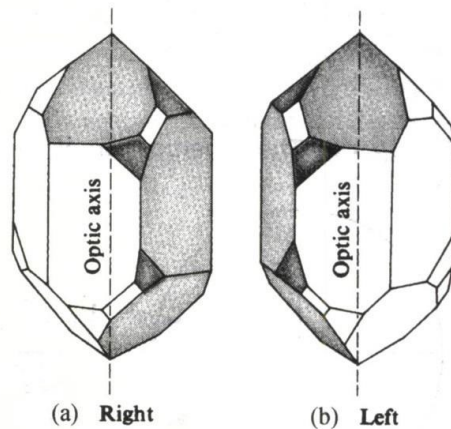
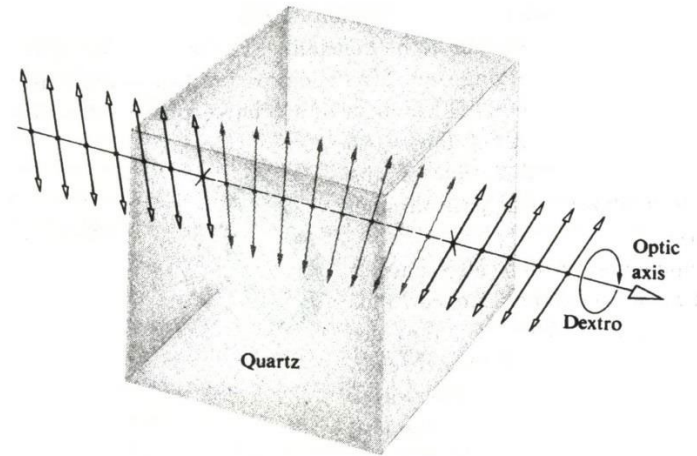


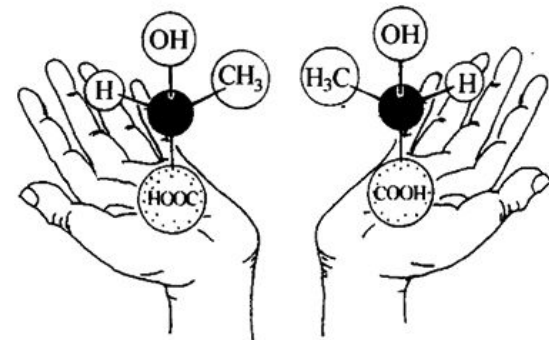
7.10 Optical Activity

Any material that causes the E-field of an incident linear plane wave to appear to rotate is said to be **optically active**.

- ❑ If the plane-of-vibration appears to have revolved clockwise: dextrorotatory or d-rotatory
- ❑ If it displaces counterclockwise: the materials is levorotatory or l-rotatory.



手性分子 chiral molecule 具有手性的分子，一般在一个C原子上按四面体方式连



接上4个互不相同的基团形成手性分子，如图示出左右两只手托出两个呈镜像关系的乳酸分子。一切具有螺旋结构的分子都是手性分子。[P]

Model: the incident linear wave is represented as a superposition of R- and L- states, and two forms of circular light propagate at different speed (circular birefringence); In traversing an optically active specimen, the two circular waves would get out-of-phase, the resultant linear wave would appear to have rotated.

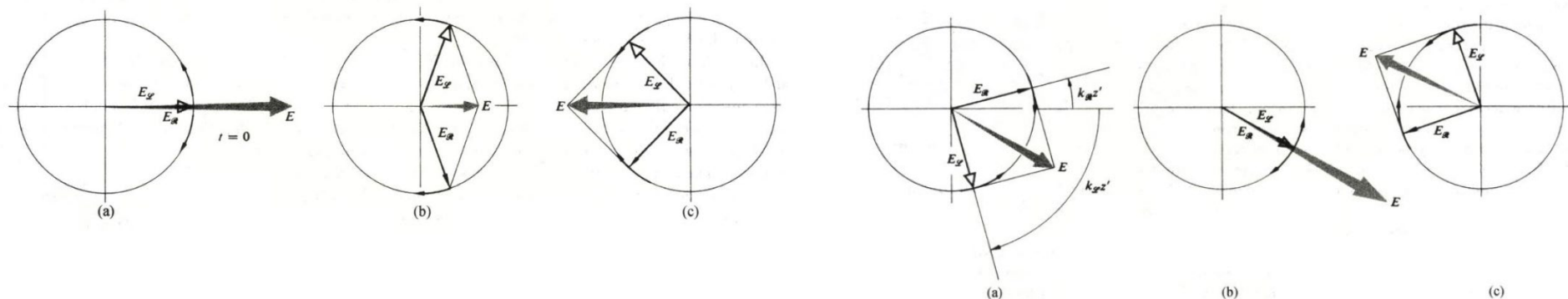
$$E_R = \frac{E_0}{2} [\hat{i} \cos(k_R z - \omega t) + \hat{j} \sin(k_R z - \omega t)]$$

$$E_L = \frac{E_0}{2} [\hat{i} \cos(k_L z - \omega t) - \hat{j} \sin(k_L z - \omega t)]$$

The resultant disturbance:

$$E = E_0 \cos[(k_R + k_L)z/2 - \omega t] \left[\hat{i} \cos \frac{(k_R - k_L)z}{2} + \hat{j} \sin \frac{(k_R - k_L)z}{2} \right]$$

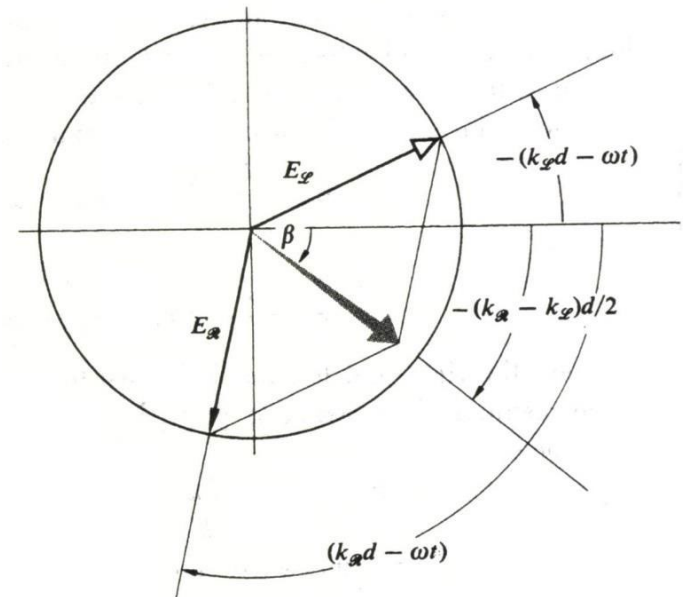
The superposition of a R- and an L- state at $z=0$ The superposition of a R- and an L- state at $z=z'$



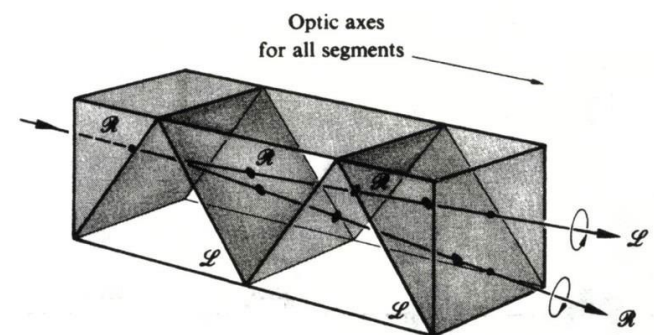
At point z , there is an angle $\beta = -\frac{(k_R - k_L)z}{2}$ with respect to its original orientation.

If the medium has a thickness d , the angle through which the plane-of-vibration rotates is:

$$\beta = \frac{\pi d}{\lambda_0} (n_L - n_R)$$



Separate the constituent R- and L- states of a linear beam using the composite prism



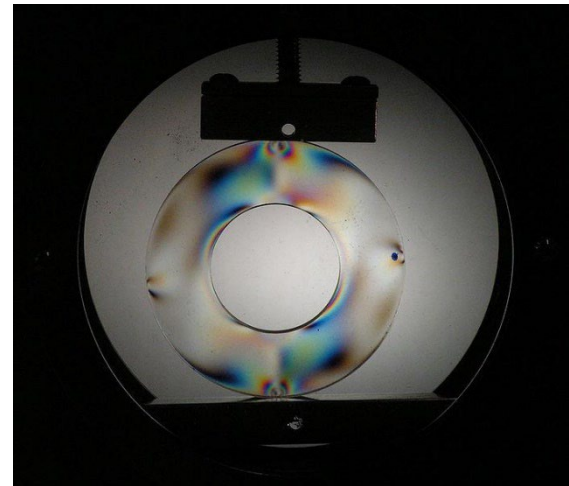
7.11 Induced Optical Effect-Optical Modulators

I. Photoelasticity (mechanical birefringence or stress birefringence)

- ❑ Normally transparent isotropic substance could be made optically anisotropic by the application of mechanical stress.
- ❑ The effective optic axis is in the direction of the stress and the induced birefringence is proportional to the stress.

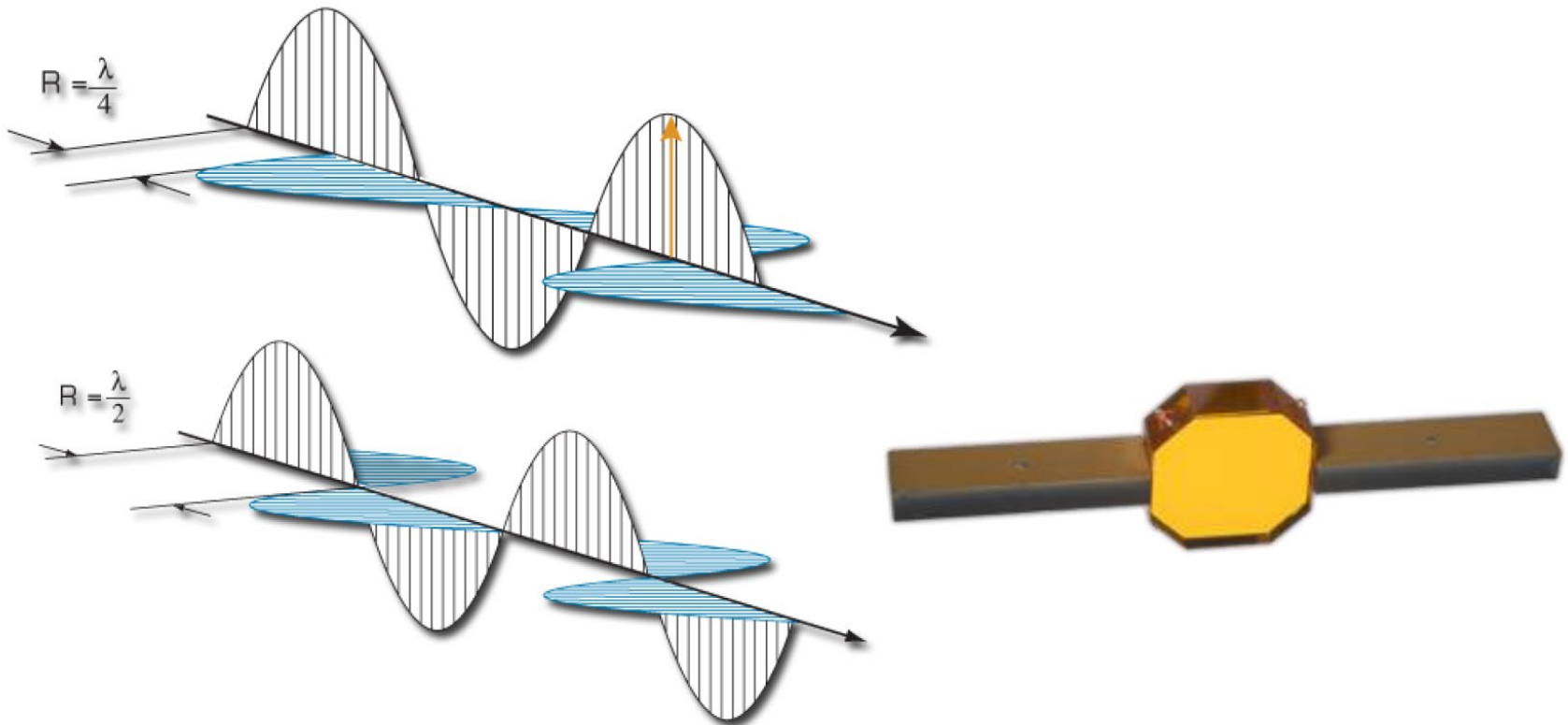
Example: a plate under vertical tension.

- ❑ σ_1 : the maximum principle stress in the vertical direction.
- ❑ σ_2 : the minimum principal stress horizontally. It is zero.
- ❑ The retardance $\propto$ the principal stress difference ($\sigma_1 - \sigma_2$)



Photoelastic Modulation

Principle : a sample of transparent solid material is stressed by compression or stretching, the material becomes birefringent, that is, different linear polarizations of light have slightly different speeds of light when passing through the material.



II. The Faraday Effect

Phenomenon: the plane-of-vibration of linear light incident on a piece of glass rotated when a strong magnetic field was applied in the propagation direction. (magneto-optic effect)

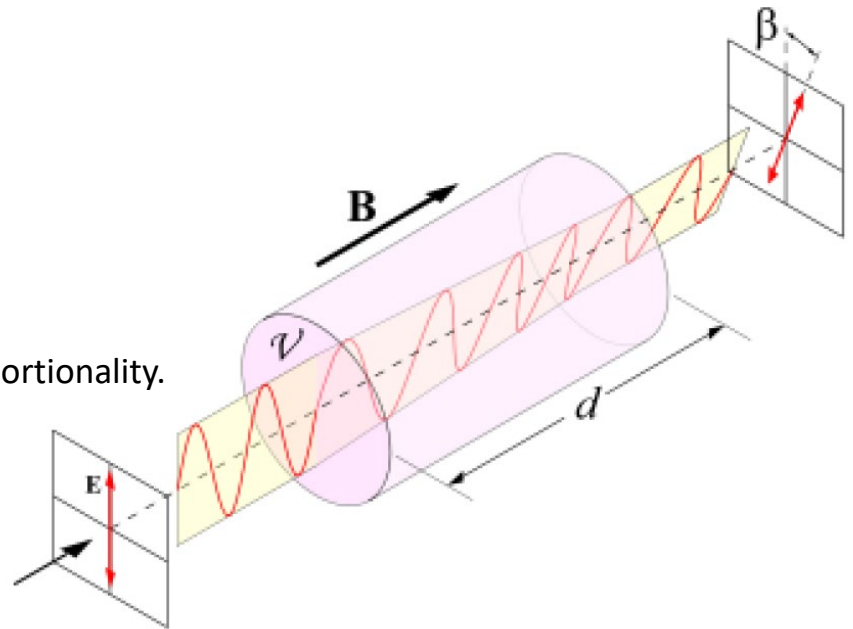
Empirical expression:

$$\beta = VBd$$

B: the static magnetic flux density;

d: the length of medium traversed;

V [Verdet constant (费尔德常数)]: a factor of proportionality.



Positive V: l-rotatory when the light moves parallel to B; d-rotatory when it propagates antiparallel to B.

Note: no such reversal of handedness occurs in the case of natural optical activity.

Classical argument for nonmagnetic materials:

Consider the incident light to be circular and monochromatic.

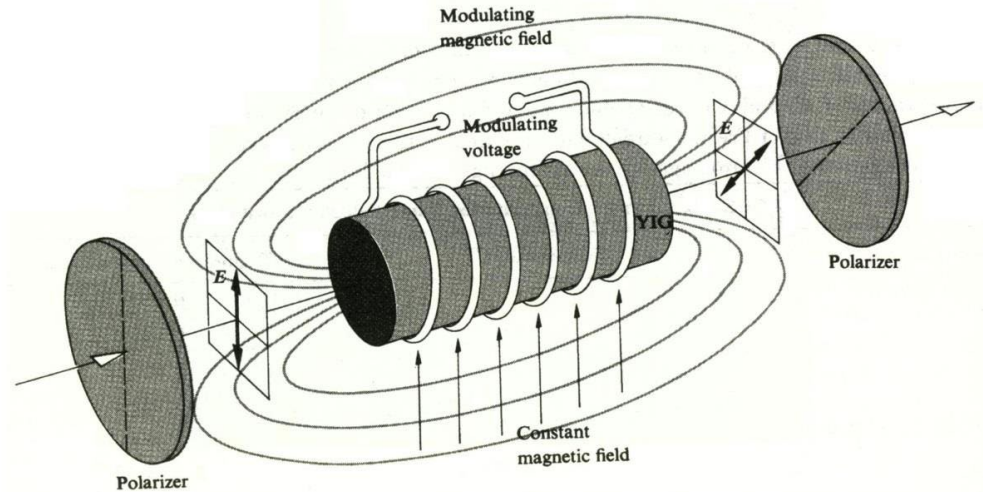
- ❑ An elastically bound electron will take on a steady-state circular orbit being driven by the rotating E-field of the wave.
- ❑ The introduction of a magnetic field perpendicular to the plane of the orbit will result in a radial force F_m on the electron.
- ❑ Its direction is determined by the handedness of the light and the direction of the B-field.
- ❑ The total radial force has two values, so two radii of the orbit.

For a given magnetic field, there will be two possible values of the electric dipole moment, the polarization, the permittivity, as well two values of the index of refraction, n_R and n_L .

Application of Faraday Effect

Optical modulators

Materials: synthetic magnetic crystal yttrium-iron garnet (YIG, 钇铁石榴石), which is used in infrared light.



Other magneto-optic effects

Voigt (in vapors) and Cotton-Mouton (in liquids) Effects: both arise when a constant magnetic field is applied to a transparent medium perpendicular to the direction of propagation of the incident light beam.

Principle: medium displays birefringence similar to that of a uniaxial crystal whose optic axis is in the direction of the dc magnetic field.

The difference of indices of refraction is proportional to the square of the applied magnetic field.

III. The Kerr Effect

Phenomenon: an isotropic transparent substance becomes birefringent when placed in an electric field E . (electro-optic effect)

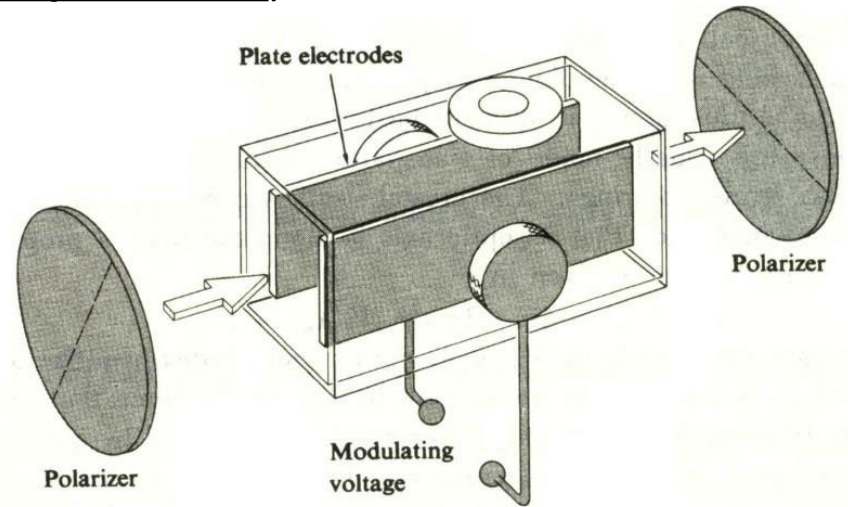
Reason: a partial alignment of anisotropic molecules by the E -field in liquids

The medium takes on the characteristics of a uniaxial crystal whose optic axis corresponds to the direction of the applied field.

The difference of two indices (parallel and perpendicular to the E -field) $\Delta n = n_{\parallel} - n_{\perp}$:

$$\Delta n = \lambda_0 K E^2$$

K : the Kerr constant.



Kerr shutter or optical modulator:

The retardation

$$\Delta\phi = 2\pi K l V^2 / d^2$$

l : the effective length;

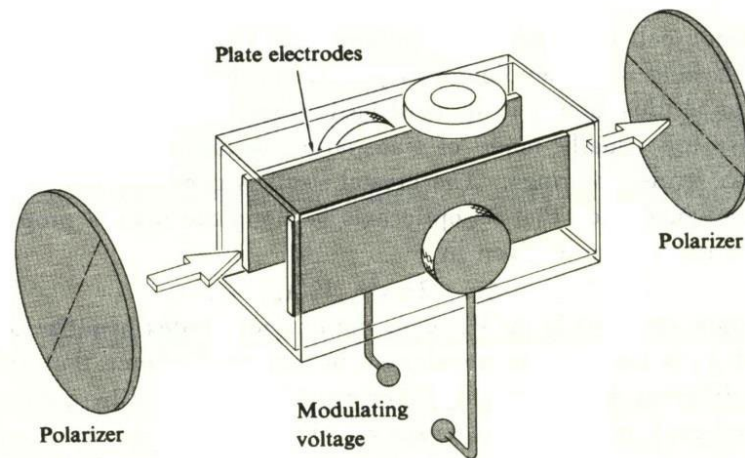
d : the distance of separation

For example: a nitrobenzene cell with $d=1\text{cm}$, l is several cm.

A half-wave plate: $V=3\times 10^4\text{V}$

Respond frequency: up to 10^{10}Hz

Application: shutters in high-speed photography; light-beam choppers



Homework

□ 8.45, 8.46, 8.49